

EduGenie AI Study Hub

Full-Stack Technical Project Documentation & System Architecture

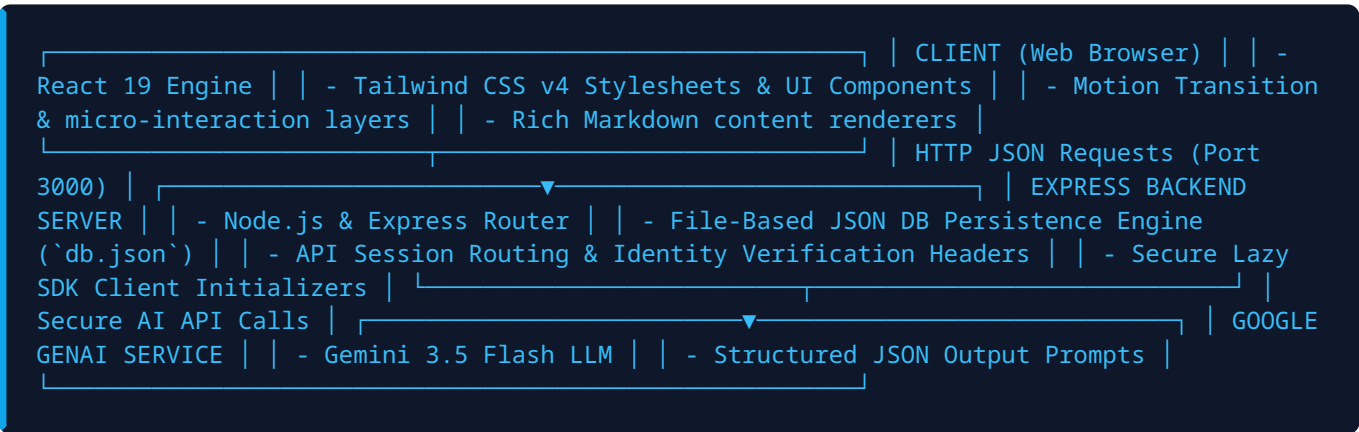
1. Executive Summary

EduGenie is a fully responsive, modern full-stack educational companion designed to empower students and teachers. It integrates generative AI to translate complex study materials into digestible knowledge structures, interactive assessments, customized learning roadmaps, and formatted study newsletters.

The core philosophy of EduGenie centers on **Architectural Security & High-Fidelity UI/UX**, featuring a server-side proxy layer that shields raw Gemini API keys from client browsers and a Tailwind-powered Single Page Application (SPA) designed with spacious grids, smooth motion transitions, and customizable templates.

2. System Architecture

EduGenie is designed as a Full-Stack Decoupled Application hosted within a Node.js container environment. The architecture isolates the client layer from direct AI platform calls to eliminate credential leakage vectors.



Architectural Key Characteristics

- **Vite Dev Middleware:** In development, Express mounts Vite dynamically to hot-serve client assets, providing high-performance local bundling on port 3000.
- **Production CJS Bundling:** In production, Vite compiles client static assets to the `/dist` directory, while esbuild bundles the Express server file (`server.ts`) into a standalone CommonJS file (`dist/server.cjs`) to prevent ES-Module runtime compatibility issues.
- **Strict Security Isolation:** The client has zero awareness of the `@google/genai` SDK or raw Gemini API credentials. All AI operations are proxied via server routes (e.g., `/api/recommendations`, etc.), preventing client-side API key leaks.

3. Technology Stack & Dependencies

Frontend Core Frameworks

- **React 19:** Powered by functional components, hook states, and optimized effect synchronization.
- **Vite 6:** Next-generation ultra-fast builder and assets packager.
- **Tailwind CSS v4:** Utility-first CSS compiling via native `@import "tailwindcss";` configurations.
- **Motion:** Modern react animation library providing high-fidelity, hardware-accelerated viewport transitions.
- **Lucide React:** Premium icon toolkit utilizing clean vector SVGs.
- **React-Markdown:** Safely converts unstructured text blocks and mathematical models into clean HTML layouts.

Backend Core Frameworks

- **Express 4:** High-performance HTTP server router handling data flow.
- **TSX:** High-performance TypeScript executor for development.
- **Esbuild:** Fast bundler producing production-grade `dist/server.cjs`.
- **Google GenAI SDK:** Integrates modern `@google/genai` TypeScript patterns for model orchestration.

4. Database Schema (db.json)

The application implements structured key-value file persistence in `db.json`, functioning as a reliable local relational store.

```
{
  "users": [
    {
      "user_id": "string (Primary Key)",
      "name": "string",
      "email": "string (Unique)",
      "password": "string (Hashed/Protected)",
      "created_at": "ISO-8601 Timestamp"
    }
  ],
  "queries": [
    {
      "query_id": "string",
      "user_id": "string (Foreign Key -> users.user_id)",
      "query_text": "string",
      "query_type": "Q&A | Explain | Quiz | Summary | Roadmap",
      "created_at": "ISO-8601 Timestamp"
    }
  ],
  "quizzes": [
```

```

{
  "quiz_id": "string",
  "query_id": "string (Foreign Key -> queries.query_id)",
  "question_text": "string",
  "option_a": "string",
  "option_b": "string",
  "option_c": "string",
  "option_d": "string",
  "correct_answer": "A | B | C | D"
}
],
"emails": [
  {
    "email_id": "string",
    "user_id": "string (Foreign Key -> users.user_id)",
    "sender": "string (e.g. EduGenie AI Assistant)",
    "recipient": "string",
    "subject": "string",
    "body_html": "string (Structured HTML format)",
    "body_markdown": "string (Markdown format)",
    "theme": "academic | emerald | midnight | minimalist",
    "created_at": "ISO-8601 Timestamp"
  }
]
}

```

5. System Features & Modules

1. Unified Authentication Studio

Purpose: Manages localized profiles, user registration, and secure credentials pairing.

Features: Beautiful transition animations, visual validation warnings, and auto-seeding for a built-in interactive demo sandbox (student@edugenie.local).

2. Q&A Core Engine

Purpose: Delivers in-depth explanations on complex academic queries.

Features: Smart topic classification, markdown-friendly answers, and integrated history hooks.

3. Explain Studio (Adaptive Cognitive Scaling)

Purpose: Rewrites any academic topic to match different cognitive levels.

Cognitive Levels Supported:

- **Analogy Mode:** Connects abstract concepts to daily examples.
- **5-Year Old:** Simplifies vocabulary and minimizes cognitive overhead.
- **High School:** Balances core scientific facts with readable structures.
- **Graduate Level:** Delivers formal, rigorous academic and mathematical explanations.

4. Quiz Master (Interactive MCQs Generator)

Purpose: Creates automated multiple-choice test sets from a single prompt.

Features: Interactive feedback on correct/incorrect selections, cumulative scorecard calculations, and instant generation options.

5. Summary Studio

Purpose: Extracts bullet summaries, action items, and structural glossaries from massive raw texts.

6. Study Roadmaps

Purpose: Generates dynamic visual timelines containing chronological study milestones and resources.

7. EduMail Studio (Newsletter Compiler & Cloner)

Purpose: Converts generated summaries, roadmaps, or quiz scorecards into highly styled HTML emails.

Visual Themes Available:

- **Classic Academic:** Elegant royal indigo headers, clear serif fonts, and classic borders.
- **Emerald Leaf:** Forest green visual cues, organic spacing, and mint-themed table lines.
- **Midnight Study:** Slate-dark layout with cyan neon glowing metric highlights.
- **Minimal Memo:** Brutalist typewriter spacing with high contrast borders and monospace fonts.

Functional Duplication Engine: Includes full mock-outbox transmission logs, live viewport frame rendering (Desktop vs. Mobile), and duplication parameters allowing users to re-send, forward, or duplicate past emails with a single click.

6. API Endpoint Inventory

Endpoint	Method	Security Header	Request Payload	Description
/api/auth/register	POST	None	{ name, email, password }	Registers a new student profile
/api/auth/login	POST	None	{ email, password }	Validates credentials and returns session variables
/api/history	GET	x-user-id	None	Retrieves all logs, queries, and linked metrics
/api/mail/send	POST	x-user-id	{ recipient, subject, body_markdown, body_html, theme }	Simulates SMTP delivery, stores email in db logs
/api/mail/history	GET	x-user-id	None	Fetches all outgoing mail logs for the current profile
/api/mail/delete	POST	x-user-id	{ email_id }	Deletes a stored email from outbox history
/api/mail/duplicate	POST	x-user-id	{ email_id, new_recipient }	Duplicates, clones, and re-transmits an existing study log

7. Configuration & Environment Setup

All secret environment variables are isolated inside server-side configurations. Create a `.env` file in the root directory:

```
# .env.example
PORT=3000
NODE_ENV=production
GEMINI_API_KEY=your_gemini_api_key_here
```

8. Build, Execution & Deployment Lifecycle

EduGenie uses highly optimized execution scripts defined in `package.json` to manage development, compilation, and production start commands:

Development Environment:

Launches the full-stack server using fast-compiling type stripping.

```
npm run dev
```

Static Asset Linter:

Validates syntax errors and type safety before compile cycles.

```
npm run lint
```

Production Bundling Phase:

Compiles and packages Vite static files into `dist/`, and bundles the Express TypeScript codebase (`server.ts`) into a streamlined CommonJS package at `dist/server.cjs` via esbuild.

```
npm run build
```